THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE FULL SYLLABUS PAPER SET 10

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The colour of turmeric paper in an acidic solution: 1

(a) stays yellow **(b)** turns red **(c)** turns green **(d)** turns blue

2. Factory waste that is acidic is treated with: 1

(a) basic substances **(b)** sugar **(c)** acids **(d)** salt

3. A device used to open or close a circuit is a: 1

(a) filament **(b)** fuse only **(c)** terminal **(d)** switch

4. The path along which electric current flows is called an electric: 1

(a) cell **(b)** filament **(c)** conductor **(d)** circuit

5. The soft metal that can be cut with a knife is: 1

(a) copper **(b)** iron **(c)** gold **(d)** sodium

6. The green coating on copper vessels is due to: 1

(a) rusting of iron **(b)** photosynthesis **(c)** evaporation **(d)** corrosion

7. Ripening of a fruit is a: 1

(a) chemical change **(b)** physical change **(c)** freezing **(d)** melting

8. Burning magnesium ribbon gives: 1

(a) carbon dioxide **(b)** magnesium oxide
(c) magnesium chloride **(d)** magnesium hydroxide

9. The bump in the throat of adolescent boys is called the: 1

(a) voice string **(b)** windpipe **(c)** Adam's apple **(d)** tonsil

10. Which of these is a good social-media habit? 1

(a) Think before posting **(b)** Share passwords
(c) Share personal photos with strangers **(d)** Spread rumours

11. A fan helps us to feel cool because it: 1

(a) stops evaporation **(b)** cools the air
(c) speeds up evaporation of sweat **(d)** makes air colder

12. Which of the following transfers heat without needing a material medium? 1

(a) Convection **(b)** Evaporation **(c)** Conduction **(d)** Radiation

13. 72 km/h is equal to: 1

(a) 72 m/s **(b)** 1.2 m/s **(c)** 20 m/s **(d)** 259.2 m/s

14. A bus covers 360 km in 6 hours. Its average speed is: 1

(a) 70 km/h **(b)** 2160 km/h **(c)** 60 km/h **(d)** 30 km/h

15. The air passes from the nostrils into the lungs through the: 1

(a) oesophagus **(b)** windpipe **(c)** rectum **(d)** stomach

16. The food pipe is called the: 1

(a) oesophagus **(b)** duodenum **(c)** rectum **(d)** windpipe

17. During photosynthesis, light energy is changed into: 1

(a) electrical energy **(b)** sound

(c) heat only **(d)** chemical energy

18. Photosynthesis occurs mainly in the: 1

(a) root cap **(b)** xylem **(c)** chloroplasts **(d)** nucleus

19. If you walk towards a plane mirror at 1 m/s, your image moves towards you at: 1

(a) 1 m/s (so closing speed is 2 m/s) **(b)** 0 m/s

(c) 0.5 m/s **(d)** 4 m/s

20. A lunar eclipse occurs on a: 1

(a) quarter moon **(b)** new moon day **(c)** every day **(d)** full moon day

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 15–25 words each.

21. What is neutralisation? Give an example. 2

22. Why are electrical wires coated with plastic? 2

23. What is rusting? What conditions are needed? 2

24. What are secondary sexual characteristics? Give two examples. 2

25. Why do we wear woollen clothes in winter? 2

26. What is a clepsydra? 2

27. What is the role of saliva? 2

28. What is photosynthesis? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. State two uses of conductors and two of insulators. 3

30. Explain how a change in state is a physical change. 3

31. Why is a black-painted pot good for solar cookers? 3

32. Describe the digestive system of a ruminant. 3

33. Why does a dark room seem bright when a torch is switched on and a mirror used? 3

34. Why is a lunar eclipse safe to watch? 3

35. State two uses of bases in daily life. 3

OR

What is the role of acid rain in damaging monuments? 3

36. How do leaves exchange gases? 3

OR

Give reasons: 3
(a) leaves are green;
(b) root hairs are thin.

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 80–120 words each.

37. **Case study:** A ship's hull is rusting due to salty water. 5
(i) Why does sea water speed it up?
(ii) What coating?
(iii) Which metal can be fixed to it?
(iv) What is that protection called?

OR

(i) Describe malleability and ductility. 5
(ii) Name two metals for each.
(iii) Which is the most malleable?

38. **Case study:** Meera feels irritable before exams and is anxious about her appearance. 5
(i) Is it normal?
(ii) Name the cause.
(iii) Suggest two remedies.
(iv) When to seek help?

OR

(i) Explain the importance of iron and calcium. 5
(ii) Give sources.
(iii) Disorders from their deficiency.

39. (i) State two uses of knowing speed in daily life. 5
(ii) Give a numerical example.
(iii) Explain why units matter.

OR

(i) Draw (describe) a distance–time graph for a bus stopping at a station. 5
(ii) Explain each part.

40. (i) List the properties of light. 5
(ii) Explain two with activities.
(iii) Importance.

OR

(i) How does the shadow of a person change from morning to noon? 5
(ii) Why?
(iii) Use in sundials.

END OF QUESTION PAPER Total: 80 Marks

MASTERMIND